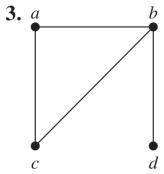
Mat2540 HW 10

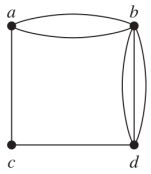
For Exercises 3–9, determine whether the graph shown has directed or undirected edges, whether it has multiple edges, and whether it has one or more loops. Use your answers to determine the type of graph in Table 1 this graph is.

3.



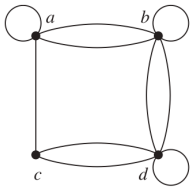
undirected edge
no multiple edges ⇒ simple graph
no loop

4.



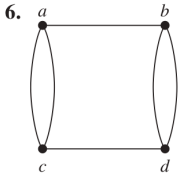
undirected edge
multiple edges ⇒ multigraph
no loop

5.



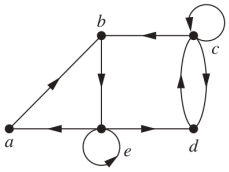
undirected edge
multiple edges ⇒ pseudograph
loops

6.



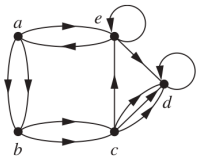
undirected edge
multiple edges ⇒ multigraph
no loop

7.



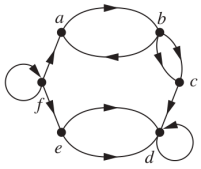
directed edge
multiple edges ⇒ directed multigraph
loops

8.



directed edge
multiple edges ⇒ directed multigraph
loops

9.



directed edge
multiple edges ⇒ directed multigraph
loops

29. Describe a graph model that represents whether each person at a party knows the name of each other person at the party. Should the edges be directed or undirected? Should multiple edges be allowed? Should loops be allowed?

We can build a simple graph to describe this social network model. If two people know each other, then there is an undirected edge between the two vertices which represent these two people.

Thus, it is an undirected graph, no multiple edges needed and no loops.

30. Describe a graph model that represents a subway system in a large city. Should edges be directed or undirected? Should multiple edges be allowed? Should loops be allowed?

Generally speaking, most subways go both ways and an undirected graph should be good enough to describe it.

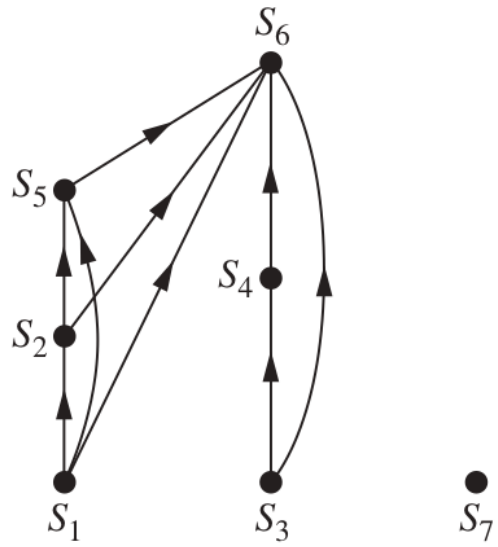
If not, we will need 2 directed edges between two stations A, B, one edge goes from A to B, the other goes from B to A.

And loop is allowed since some terminal station need to turn the train around and let it leave the station.

31. For each course at a university, there may be one or more other courses that are its prerequisites. How can a graph be used to model these courses and which courses are prerequisites for which courses? Should edges be directed or undirected? Looking at the graph model, how can we find courses that do not have any prerequisites and how can we find courses that are not the prerequisite for any other courses?

If course A is a prerequisite of course B, then there is a directed edge from A to B.
We will need a directed simple graph to describe this model. Under this setup,
the course without any prerequisites has zero out-degree. and the couse which is not a prerequisite for any other courses has zero in-degree.

35. Construct a precedence graph for the following program:

S_1: $x := 0$
S_2: $x := x + 1$
S_3: $y := 2$
S_4: $z := y$
S_5: $x := x + 2$
S_6: $y := x + z$
S_7: $z := 4$

35.